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BICOL UNIVERSITY
POLANGUI
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BAGONG PILIPINAS

IT 112: Web Systems and Technologies
BSIT- 2A 2nd Semester
S.Y. 2025-2026

Final Project – Phase 4: Form Submission and Data Insertion
Combat Coders: The End of Bugs Reign

Submitted by:

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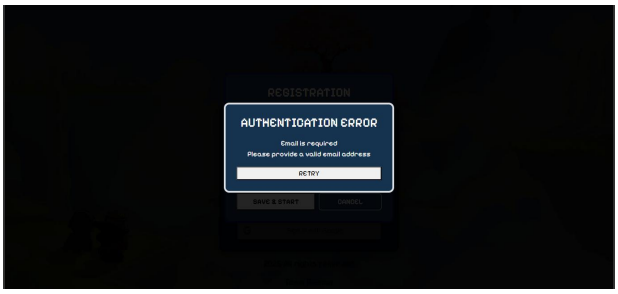
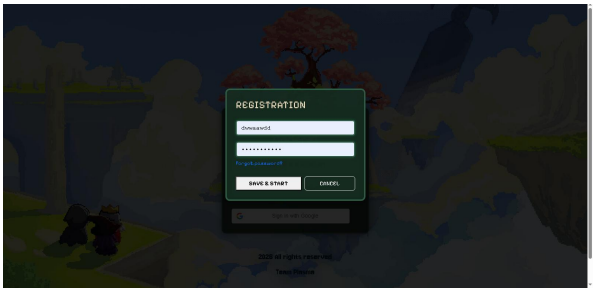
Introduction

This phase focuses on integrating the frontend with the backend by connecting HTML forms to API endpoints using JavaScript. The goal is to enable real-time data submission and ensure that user inputs are properly validated, processed, and stored in the database. Through this process, the system becomes fully functional by allowing interaction between the user interface and the server, completing the core workflow from data input to data storage.

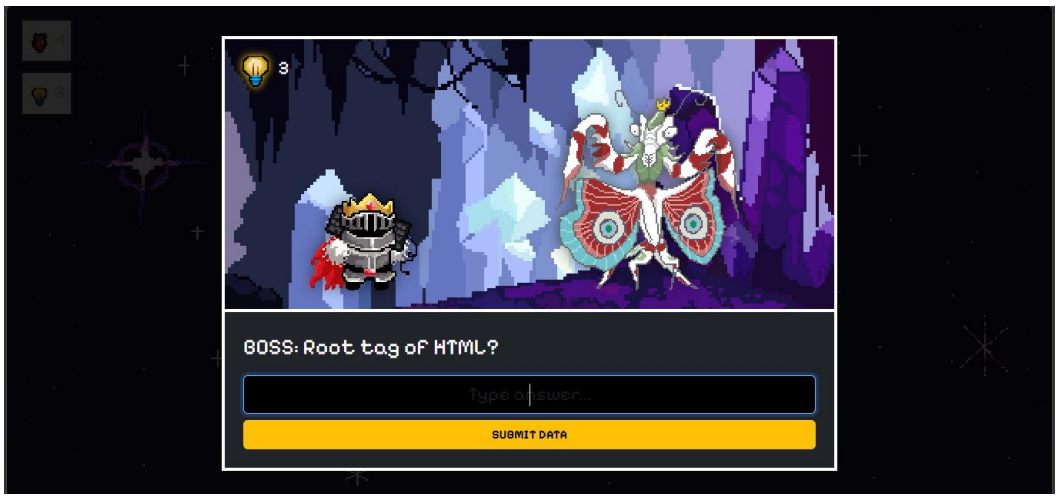
I. Screenshot of the HTML form interface

A. Text Fields





B. Input



II. Screenshot of JS code used for API connection

The following screenshots present the JavaScript code used to connect the frontend forms to the backend API. The implementation includes configuration of the base API URL, reusable request functions using fetch(), and form handling logic for user registration and login. Input validation, error handling, and response processing are also demonstrated to ensure proper communication between the client-side interface and the server.

final-project-Team_Plasma-BSIT2a > frontend > js > api-connection.js

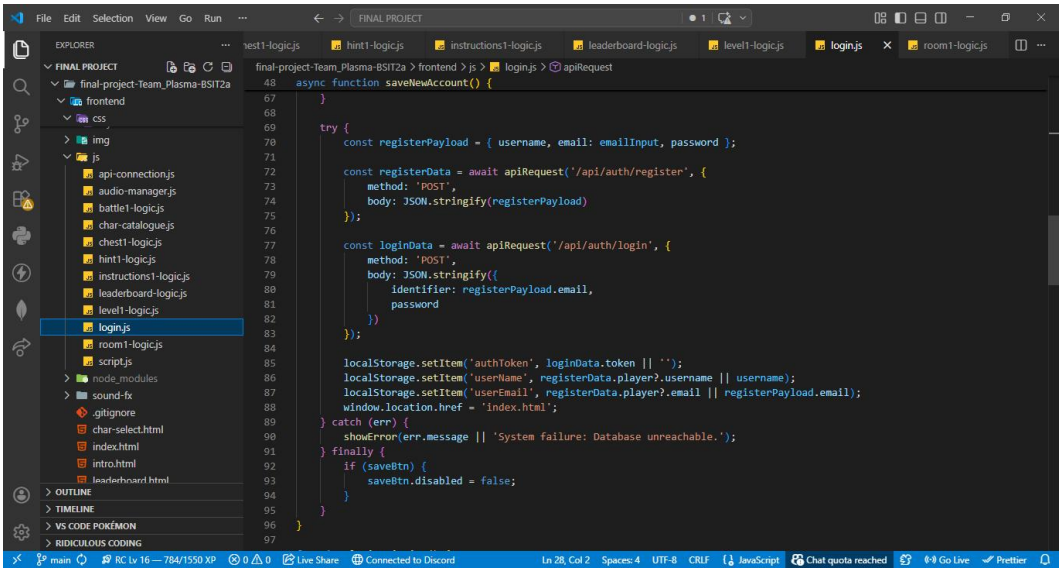
```javascript
window.API_BASE_URL = window.API_BASE_URL || 'http://localhost:5000';
```

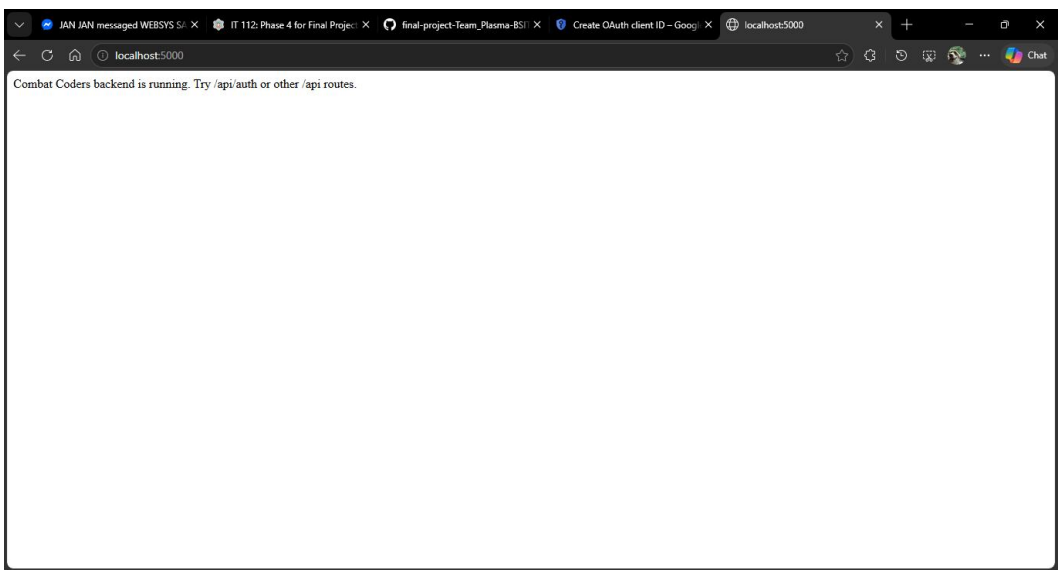
final-project-Team_Plasma-BSIT2a > frontend > js > login.js > apiRequest

```javascript
const API_BASE = window.API_BASE_URL || 'http://localhost:5000';

function usernameToEmail(username) {
    const safe = String(username || '').trim().toLowerCase().replace(/[^a-z0-9._-]/g, '');
    return `${safe || 'player'}@combatcoders.local`;
}

async function apiRequest(path, options = {}) {
    let response;
    try {
        response = await fetch(`${API_BASE}${path}`, {
            headers: { 'Content-Type': 'application/json', ...(options.headers || {}) },
            credentials: 'include',
            ...options
        });
    } catch (networkErr) {
        throw new Error(networkErr?.message || 'Network error: unable to reach backend');
    }

    const data = await response.json().catch(() => ({}));
    if (!response.ok) {
        const validation = Array.isArray(data?.errors)
            ? data.errors.map((e) => e?.message).filter(Boolean).join('\n')
            : '';
        throw new Error(validation || data.message || `Request failed (HTTP ${response.status})`);
    }
    return data;
}

function openRegModal() { document.getElementById('regModal').classList.remove('d-none'); }
function closeRegModal() { document.getElementById('regModal').classList.add('d-none'); }
function closeErrorModal() { document.getElementById('errorModal').classList.add('d-none'); }
```

final-project-Team_Plasma-BSIT2a > frontend > js > login.js > apiRequest

```javascript
async function saveNewAccount() {
    const username = document.getElementById('regName').value.trim();
    const emailInput = document.getElementById('regEmail')?.value.trim();
    const password = document.getElementById('regPass').value.trim();

    if (!username || !emailInput || !password) {
        showError('Data required: Entry fields are empty.');
        return;
    }

    if (!isValidEmail(emailInput)) {
        showError('Please provide a valid email address.');
        return;
    }

    // Prevent accidental double-click spamming the auth endpoints (can trigger 429).
    const saveBtn = document.querySelector('#regModal button[onclick="saveNewAccount()"]');
    if (saveBtn) {
        saveBtn.disabled = true;
    }

    try {
        const registerPayload = { username, email: emailInput, password };

        const registerData = await apiRequest('/api/auth/register', {
            method: 'POST',
            body: JSON.stringify(registerPayload)
        });

        const loginData = await apiRequest('/api/auth/login', {
            method: 'POST',
            body: JSON.stringify(
```





III. Screenshot of MongoDB document inserted

The following screenshots display the successfully inserted documents in the MongoDB database. These records confirm that data submitted from the frontend forms is properly processed by the backend and stored in their respective collections. Each document reflects the defined schema structure, demonstrating correct data flow and successful integration between the frontend, backend, and database.

CLUSTERS (1)

Search clusters

- CombatCoders
  - admin
  - local
  - sample_mflix
  - test
    - attempts
    - battles
    - bosses
    - chests
    - enemies
    - finalbattles
    - hearts

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df6c')
battle_id : ObjectId('69e4e1d509e457534d29df65')
player_id : ObjectId('69e4e1d509e457534d29df62')
input_answer : "console.log('hello')"
is_correct : true
hearts_lost : 0
__v : 0
```

CLUSTERS (1)

Search clusters

- CombatCoders
  - admin
  - local
  - sample_mflix
  - test
    - attempts
    - battles
    - bosses
    - chests
    - enemies
    - finalbattles
    - hearts

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df65')
player_id : ObjectId('69e4e1d509e457534d29df62')
path_id : ObjectId('69e4e1d509e457534d29df63')
description : "console.log battle"
reward_hearts : 1
reward_hints : 1
__v : 0
```

CLUSTERS (1)

Search clusters

- CombatCoders
  - admin
  - local
  - sample_mflix
  - test
    - attempts
    - battles
    - bosses
    - chests
    - enemies
    - finalbattles
    - hearts

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df68')
path_id : ObjectId('69e4e1d509e457534d29df63')
__v : 0
```

CLUSTERS (1)

Search clusters

- CombatCoders
  - admin
  - local
  - sample_mflix
  - test
    - attempts
    - battles
    - bosses
    - chests
    - enemies
    - finalbattles
    - hearts

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df66')
battle_id : ObjectId('69e4e1d509e457534d29df65')
enemy_name : "Syntax Bug"
enemy_health : 100
__v : 0
```

CLUSTERS (1)

Search clusters

- bosses
- chests
- enemies
- finalbattles
- hearts
- hints
- leaderboards
- paths
- players
- progresses
- rooms
- users

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df6a')
battle_id : ObjectId('69e4e1d509e457534d29df65')
room_id : ObjectId('69e4e1d509e457534d29df64')
chest_id : ObjectId('69e4e1d509e457534d29df68')
score : 0
__v : 0
```

CLUSTERS (1)

Search clusters

- bosses
- chests
- enemies
- finalbattles
- hearts
- hints
- leaderboards
- paths
- players
- progresses
- rooms
- users

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df67')
enemy_id : ObjectId('69e4e1d509e457534d29df66')
cost : 10
__v : 0
```

CLUSTERS (1)

Search clusters

- bosses
- chests
- enemies
- finalbattles
- hearts
- **hints**
- leaderboards
- paths
- players
- progresses
- rooms
- users

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df69')
chest_id : ObjectId('69e4e1d509e457534d29df68')
hint_text : "Check your semicolons!"
cost : 5
__v : 0
```

CLUSTERS (1)

Search clusters

- bosses
- chests
- enemies
- finalbattles
- hearts
- hints
- **leaderboards**
- paths
- players
- progresses
- rooms
- users

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df6e')
player_id : ObjectId('69e4e1d509e457534d29df62')
score : 100
rank_number : 1
__v : 0
```

CLUSTERS (1)

Search clusters

- bosses
- chests
- enemies
- finalbattles
- hearts
- hints
- leaderboards
- **paths**
- players
- progresses
- rooms
- users

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df63')
path_name : "Beginner Path"
difficulty_level : 1
player_id : ObjectId('69e4e1d509e457534d29df62')
__v : 0
```

CLUSTERS (1)

Search clusters

- bosses
- chests
- enemies
- finalbattles
- hearts
- hints
- leaderboards
- paths
- **players**
- progresses
- rooms
- users

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e8934fe596aaf1d6cc2e69')
username : "user1507bb6f"
password : "$2b$10$ZhpDRS7dmLKE8r7nAW9vCOs4osiYL7ZMo7t288tb473Yomi9yGi9a"
email : "test1507bb6f@example.com"
hearts : 3
hints : 1
level : 0
createdAt : 2026-04-22T09:22:23.060+00:00
updatedAt : 2026-04-22T09:22:23.060+00:00
__v : 0
```

```
_id: ObjectId('69e89435be6d44f2979ab224')
username : "piapip"
password : "$2b$10$gduYNHtwpxMuaALit4K1/.ZBajtqOED4cYI/ajhzKzMCbV4FVgqam"
email : "sophiabalmaceda38@gmail.com"
hearts : 3
```

CLUSTERS (1)

Search clusters

- bosses
- chests
- enemies
- finalbattles
- hearts
- hints
- leaderboards
- paths
- players
- **progresses**
- rooms
- users

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df6d')
player_id : ObjectId('69e4e1d509e457534d29df62')
attempt_id : ObjectId('69e4e1d509e457534d29df6c')
completed : true
earned_hearts : 1
earned_hints : 1
__v : 0
```

CLUSTERS (1)

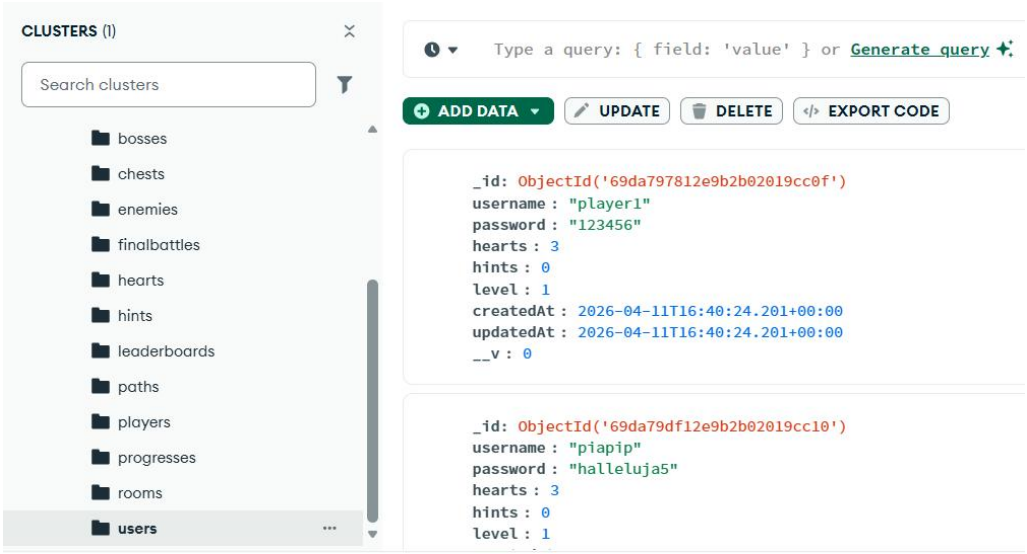
Search clusters

- bosses
- chests
- enemies
- finalbattles
- hearts
- hints
- leaderboards
- paths
- players
- progresses
- **rooms**
- users

Type a query: { field: 'value' } or Generate query

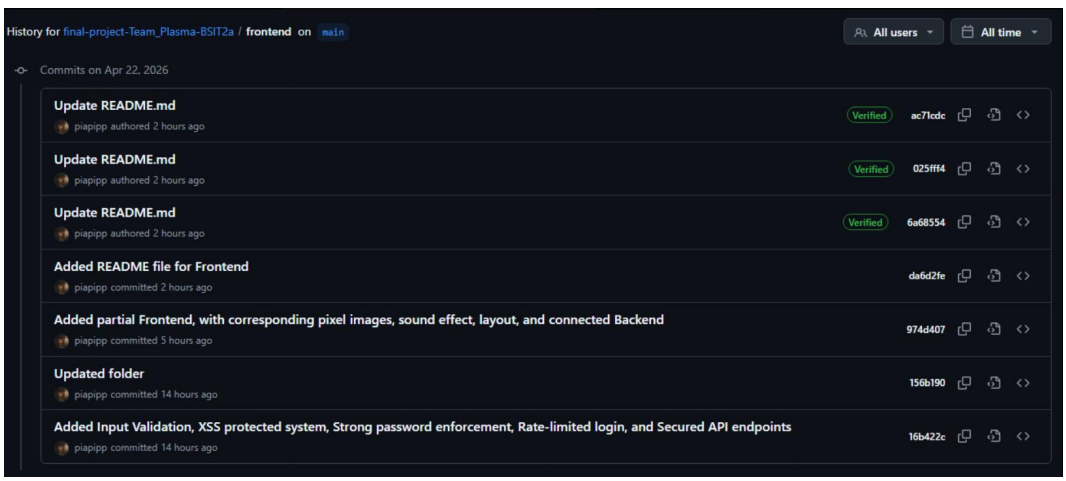
ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69e4e1d509e457534d29df64')
path_id : ObjectId('69e4e1d509e457534d29df63')
storyline : "You enter the debugging dungeon..."
__v : 0
```

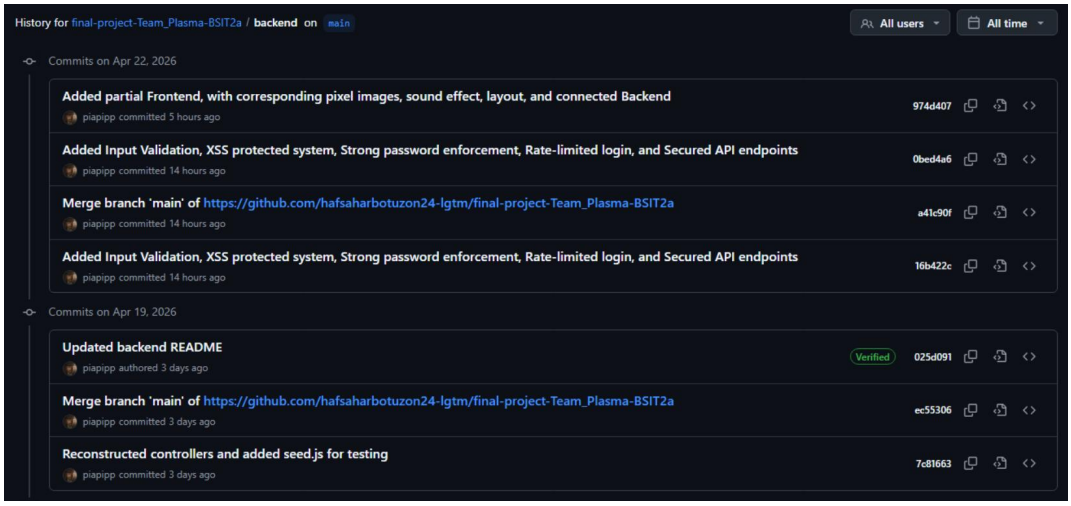


IV. Screenshot of GitHub commits (frontend and backend)

- **Frontend**



- **Backend**



V. Short reflections (2–3 sentences) from each member

Sophia Mae Alodia J. Balmaceda: *Backend Developer and Project Manager*

Created several backgrounds using Piskel. Helped with frontend tasks. Adjusted and prepared the backend beforehand for integration after the frontend was finished. Same time of the integration.

Dwayne Lester O. Diaz: *Frontend Developer*

Worked mainly on the frontend. Maintained consistent effort throughout development. Led design decisions such as pixel size and image resolution.

Joshua Cedric B. Velitario: *Database Manager*

Helped prepare the laboratory report. Developed game dialogue, including modals, scenery text, and character interactions. Created questions and challenges for enemy encounters.

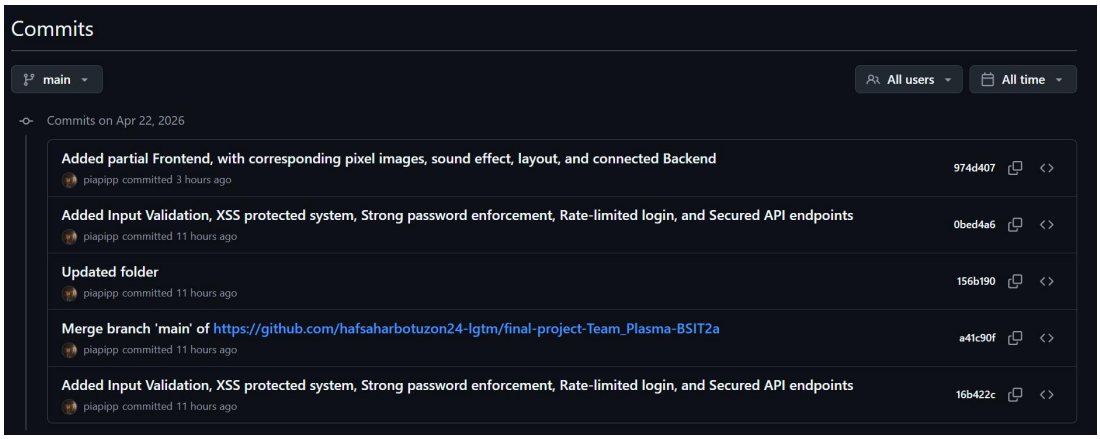
Kian Carlo O. Matutina: *Documentation Officer*

Acted as the documentation officer and captured screenshots for project records. Led the creation of the laboratory report. Made most of the questions for the different phases.

Hafsah A. Tuzon: *GitHub Manager*

Designed some backgrounds using Piskel. Created most of the game sprites, including icons and images. Helped with the laboratory report.

VI. GitHub commits



Commits on Apr 19, 2026

docs updated hafsaharbotuzon24-lgtm authored 3 days ago	Verified	9b5739c
Update README.md hafsaharbotuzon24-lgtm authored 3 days ago	Verified	848a792
Update README.md hafsaharbotuzon24-lgtm authored 3 days ago	Verified	40d6146
Update README.md hafsaharbotuzon24-lgtm authored 3 days ago	Verified	a539489
Update README.md hafsaharbotuzon24-lgtm authored 3 days ago	Verified	1cee73f
Update README.md hafsaharbotuzon24-lgtm authored 3 days ago	Verified	6edb9d8
Update README.md hafsaharbotuzon24-lgtm authored 3 days ago	Verified	dac421f
Updated backend README piapipp authored 3 days ago	Verified	025d091
Merge branch 'main' of https://github.com/hafsaharbotuzon24-lgtm/final-project-Team_Plasma-BSIT2a piapipp committed 3 days ago		ec55306
Reconstructed controllers and added seed.js for testing piapipp committed 3 days ago		7c81663

Commits

main | All users | All time

Commits on Apr 22, 2026

Merge branch 'main' of https://github.com/hafsaharbotuzon24-lgtm/final-project-Team_Plasma-BSIT2a piapipp committed 49 minutes ago		07099c3
Updated Phase 2 and 3 Lab Reports piapipp committed 49 minutes ago		41dca43
Update README.md piapipp authored 2 hours ago	Verified	ac71cdc
Update README.md piapipp authored 2 hours ago	Verified	025fff4
Update README.md piapipp authored 2 hours ago	Verified	6a68554
Added README file for Frontend piapipp committed 2 hours ago		da6d2fe
Added partial Frontend, with corresponding pixel images, sound effect, layout, and connected Backend piapipp committed 5 hours ago		974d407
Added Input Validation, XSS protected system, Strong password enforcement, Rate-limited login, and Secured API endpoints piapipp committed 14 hours ago		0bed4a6
Updated folder piapipp committed 14 hours ago		156b190
Merge branch 'main' of https://github.com/hafsaharbotuzon24-lgtm/final-project-Team_Plasma-BSIT2a piapipp committed 14 hours ago		a41c90f
Added Input Validation, XSS protected system, Strong password enforcement, Rate-limited login, and Secured API endpoints piapipp committed 14 hours ago		16b422c